



Bayes' Theorem



A hypothetical scenario

You feel sick and visit a Dr. Dr does not know what problem you have. So he runs a number of tests. After the result came back, you see that you tested positive for a very rare disease that affect 0.1% of the population.

You ask the doctor, “How certain are you that I have the disease?”

The doctor says “The test will identify 99% people correctly who have the disease. And will only incorrectly identify 1% people who do not have the disease.

Question: What are the chances that you have the disease?

Apply Bayes' Theorem

You test positive if you have the disease

Prior probability of having the disease

$P(H | E)$

$=$

$P(E | H) * P(H)$

$P(E)$

Probability of testing positive

You actually have the disease given the event which is you tested positive

The diagram illustrates Bayes' Theorem with the following components and annotations:

- Top Left:** The text "You test positive if you have the disease" has a downward arrow pointing to the term $P(E | H)$ in the numerator.
- Top Right:** The text "Prior probability of having the disease" has a downward arrow pointing to the term $P(H)$ in the numerator.
- Bottom Left:** The text "You actually have the disease given the event which is you tested positive" has an upward arrow pointing to the term $P(H | E)$ on the left side of the equation.
- Bottom Right:** The text "Probability of testing positive" has a leftward arrow pointing to the term $P(E)$ in the denominator.

Apply Bayes' Theorem

You test positive if you
have the disease

Prior probability of having
the disease

$$P(H | E) = \frac{P(E | H) * P(H)}{P(H) * P(E | H) + P(-H) * P(E | -H)}$$

You actually have the
disease given the event
which is you tested positive

Probability of testing positive



The Scenario Continues

You go to another doctor for a second opinion. The doctor also runs some test and you come positive for the disease.

Question: What is the probability you actually have the disease now?